.5 inlet parameters including allowable exhaust gas temperature (maximum

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engine exhaust manifold to the SCR chamber;

.4 catalyst block specification and arrangement in the SCR chamber;

.5 inlet parameters including allowable exhaust gas temperature (maximum and minimum) at the inlet of the SCR chamber;

.6 cross-unit parameters: allowable pressure loss (p) between inlet and outlet of SCR chamber and in the exhaust duct caused by SCR components;

.7 aspects related to the fuel oil quality resulting in continued compliance of the engine with the applicable NOx emission limit;

.8 factors related to the deterioration rate of SCR performance, e.g., exchange condition for SCR blocks and recommended exchange time of SCR blocks;

.9 controlling arrangements and settings of the SCR, e.g., model, specification of control device;

.10 measures to minimize reductant slip;

.11 parameter check method as the verification procedure: with regard to the application of the parameter check method, requirements given in paragraph 2.3.6 of the NTC 2008 and guidance given in appendix VII, paragraph 2 of the NTC 2008 should be taken into account in assessing the adequacy of a proposed procedure with analysers meeting or exceeding

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paragraph 2.3.6 of the NTC 2008 and guidance given in appendix VII, paragraph 2 of the NTC 2008 should be taken into account in assessing the adequacy of a proposed procedure with analysers meeting or exceeding the requirements of appendix III of the NTC 2008; and

.12 any other parameter(s) specified by the manufacturer.

3.3 Measures to minimize reductant slip

3.3.1 When SCR uses urea solution, ammonia solution or ammonia gas as reductant, measures to prevent reductant slip should be provided to avoid the supply of an excessive amount of reductant in the system. The reduc tant injection system should be designed to prevent emissions of any harmful substance from the system.

3.4 Pre-certification procedure

3.4.1 Test and pre-certification of an engine system fitted with SCR should be conducted either by Scheme A (as given in section 5 of these guidelines), or by Scheme B (as given in

sections 6 and 7 of these guidelines), as appropriate.
RESOLUTION MEPC.198(62)
Adopted on 15 July 2011
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CODE 2008 WITH REGARD TO PARTICULAR REQUIREMENTS RELATED TO MARINE

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sections 6 and 7 of these guidelines), as appropriate.
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CODE 2008 WITH REGARD TO PARTICULAR REQUIREMENTS RELATED TO MARINE
DIESEL ENGINES FITTED WITH SELECTIVE CATALYTIC REDUCTION (SCR) SYSTEMSMEPC 62/24
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3.5 EIAPP certificate

3.5.1 An Engine International Air Pollution Prevention (EIAPP) Certificate (see appendix I of the NTC 2008) should be issued by the Administration after approval of the Technical File.

3.5.2 When an applicant chooses the Scheme B for pre-certification, the IAPP initial survey should not be completed until the on board initial confirmation test provides compliant results. The applicant remains the responsible entity until final acceptance of the system.

4 FAMILY AND GROUP CONCEPTS FOR ENGINE SYSTEMS FITTED WITH SCR

4.1 Requirements in chapter 4 of the NTC 2008 apply equally to engine systems fitted with SCR.

5 TEST PROCEDURES FOR SCHEME A

5.1 General

5.1.1 A test for a combined system of an engine fitted with an SCR in Scheme A is to

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SOURCE: MEPC.198(62).pdf
4.1 Requirements in chapter 4 of the NTC 2008 apply equally to engine systems fitted with SCR.

5 TEST PROCEDURES FOR SCHEME A

5.1 General

5.1.1 A test for a combined system of an engine fitted with an SCR in Scheme A is to ensure compliance with the applicable NO x emission limits of MARPOL Annex VI, as required. The test bed measurement procedures of chapter 5 of the NTC 2008 should apply.

5.2 Calculation of gaseous emissions

5.2.1 The calculation method in section 5.12 of the NTC 2008 is also applied to engine systems fitted with SCR. No allowance is made for the reductant solution injected into the exhaust gas stream in respect of its effect on exhaust gas mass flow rate calculation (appendix VI) or dry/wet correction factor (equation (11), paragraph 5.12.3.2.2 of the NTC 2008). The NO x correction factor for humidity and temperature (equations (16) or (17), paragraphs 5.12.4.5 and 5.12.4.6, respectively, of the NTC 2008) should not be applied.

5.2.2 For an engine system fitted with SCR, the following parameters should be measured and recorded in the engine test report in accordance with section 5.10 of the NTC 2008:

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5.2.2 For an engine system fitted with SCR, the following parameters should be measured and recorded in the engine test report in accordance with section 5.10 of the NTC 2008:

.1 injection rate of reductant at each load point (kg/h);

.2 exhaust gas temperature at the inlet and outlet of the SCR chamber (°C);

.3 pressure loss (kPa): it is necessary to measure the pressure at inlet and at outlet of the SCR chamber and to calculate pressure loss p. If the manufacturer sets an allowable limit of p, it should be confirmed; and

.4 other parameter(s) as specified by the Administration.

6 TEST PROCEDURES FOR SCHEME B

6.1 General

6.1.1 A test for an engine system fitted with SCR in Scheme B is to ensure that the system complies with the applicable NO x emission limits in MARPOL Annex VI, as required. The test procedures in Scheme B are as follows:

.1 an engine is tested to obtain the NO x emission value (g/kWh) in accordance with paragraph 6.2.1 of these guidelines;

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with paragraph 6.2.1 of these guidelines;
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.2 the SCR NO x reduction rate may be calculated by modelling tools, taking into account geometrical reference conditions, chemical NO x conversion models as well as other parameters to be considered;

.3 an SCR chamber, not necessarily to fu ll scale, is to be tested in accordance with section 6.3 of these guidelines in order to generate data for the calculation model as that used in paragraph 6.1.1.2 of these guidelines;

.4 the NO x emission from the engine system fitted with SCR, which is calculated in accordance with section 6.4 of these guidelines using the NO x emission value from the engine and the NO x reduction rate of SCR chamber. At this point the Technical File will be completed and this NO x emission value will be entered into the supplement of the EIAPP certificate; and

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emission value from the engine and the NO x reduction rate of SCR chamber. At this point the Technical File will be completed and this NO x emission value will be entered into the supplement of the EIAPP certificate; and